Applied Economics

Regression Discontinuity Design

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Introduction

- Starting point: we are interested in the causal effect of some treatment, but subjects are not randomly allocated to treatment (no randomized experiment)
- We can use **quasi-experimental methods** to estimate a causal effect even if we cannot control for everything that is relevant. Last week: Difference-in-Differences

Today: Regression Discontinuity Design (RDD)

- can be applied when we have specific information about the rules determining treatment allocation. $\rightarrow$ exploit this to get quasi-random variation in treatment.
- not applicable everywhere, but intuitive + very credible identification assumption

RDD - Basic idea

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- Many policies/programs are based on arbitrary rules and thresholds:
  Above the threshold $\rightarrow$ in the program; below threshold $\rightarrow$ not in the program.

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- in the US, newborns weighing less than **1,500 grams** are considered to be of "very low birth weight" and receive additional treatment (Almond et al., 2010)
- in the US, the legal drinking age is **21 years old** (Carpenter and Dobkin, 2009).

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$\Rightarrow$ we can **zoom in** on a group of individuals where having been allocated to treatment is **as good as random** - hence not influenced by selection bias.

RDD - Terminology

We look for a treatment that is assigned based on being above/below a cutoff value of a continuous variable.

We call the continuous variable *running variable* because we run along it until we hit the cutoff

Running variable a: variable that determines assignment to treatment.
Cutoff/threshold c: level of the running variable above (or below) which individuals are treated.

Examples

- if a candidate gets 50.1% of the vote they're elected, 40.9% and they're not.
  $\rightarrow a =$ vote share, $c = 0.5$, treatment: getting elected into office
- if a newborn weights 1,499g, they get additional treatment, 1,500g and they don't
  $\rightarrow a =$ birth weight, $c = 1{,}500$, treatment: medical treatment

RDD - Basic Idea

At the threshold, the **probability of being treated jumps** from 0 to 1. (sharp discontinuity)

Apart from that, people right before and right after the threshold are **essentially the same**

$\Rightarrow$ Around the threshold, allocation to treatment is as good as random!

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RDD: Measure the gap in the outcome for those right before vs. after threshold to get treatment effect of interest

Examples

- How different are newborns weighing just below vs. just above 1,500g, other than access to additional medical care?
- Compute difference in later life health between individuals just above and just below the birth weight cutoff to estimate causal effect of access to medical treatment on long-run health outcomes

RDD - Caution

Gap in outcome at the threshold $= \delta =$ *local* average treatment effect (LATE)

- RDD only tells you the impact of treatment on outcome **around the cutoff value** of the running variable. (e.g. effect of getting medical treatment at 1,500g)
- Limited **external validity**: you cannot extrapolate to the entire population. (e.g. effect of getting medical treatment at 3,000g birth weight)
- RDD is a very **data-hungry method**: only works with sufficiently large samples, because we essentially throw away observations that are not close to the cutoff.

Example: Mortality Effect of Alcohol Consumption

RDD Framework - Example

We are interested in the causal effect of **alcohol consumption** by young adults on **mortality**.

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In the US, alcohol consumption is prohibited before the **age of 21**.

The cutoff age, 21, separates those who can drink legally and those who can't:

Treatment status D_a is a deterministic and discontinuous function of the running variable *age*:

$$D_a = \begin{cases} 1 & \text{if } age \geq 21 \\ 0 & \text{if } age < 21 \end{cases}$$

RDD - Graphs: Running Variable and Cutoff

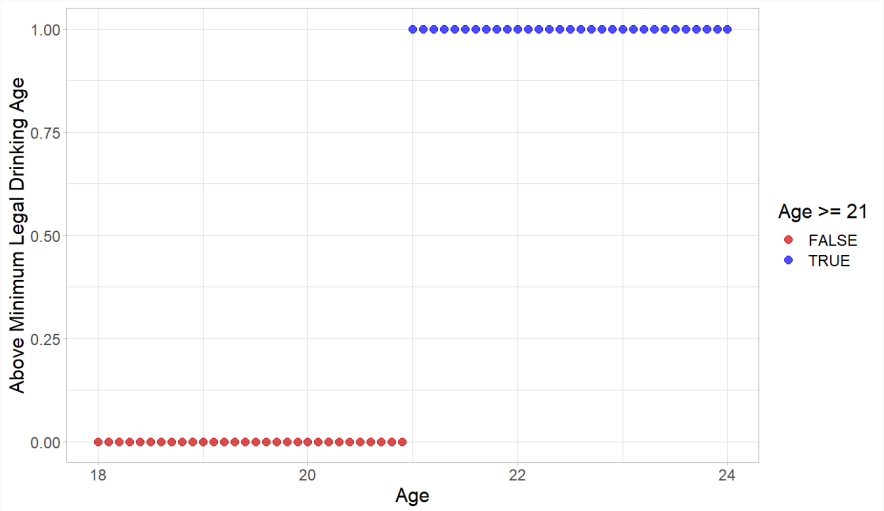
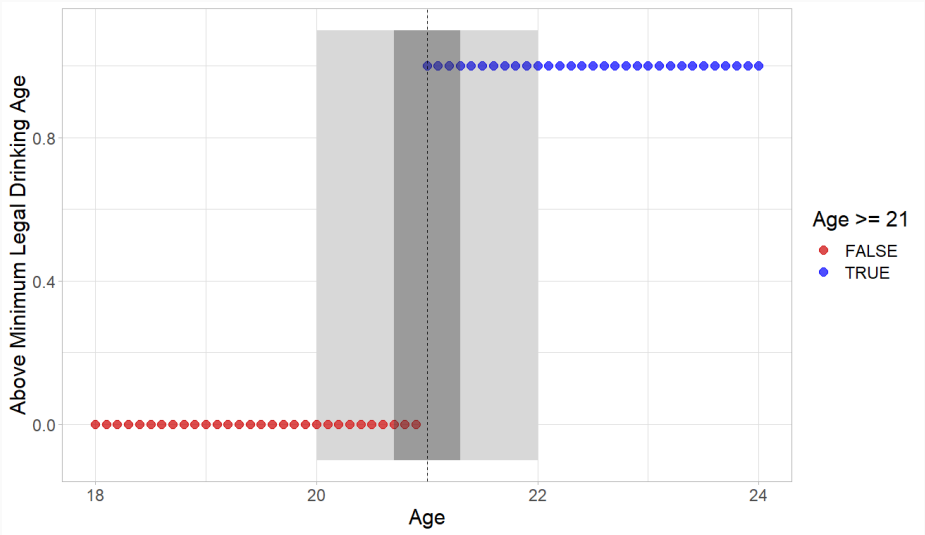
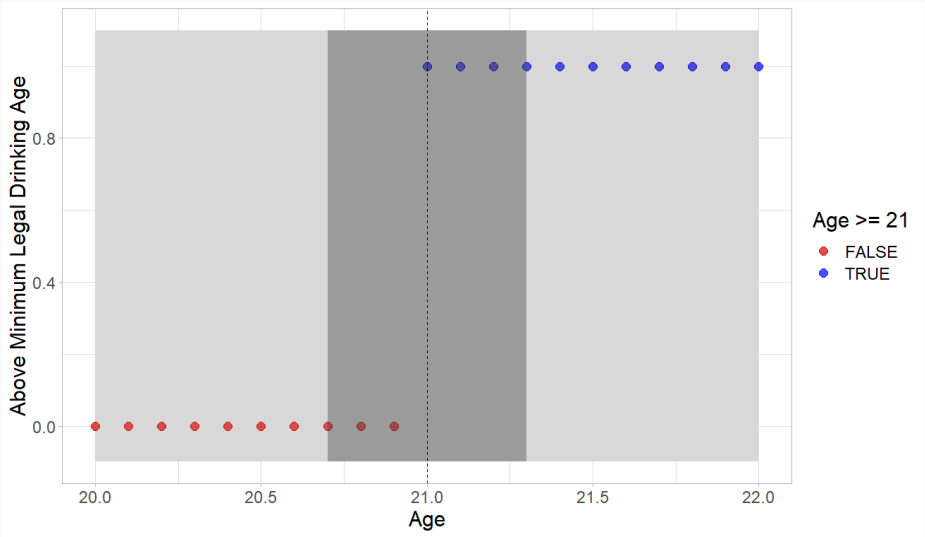


Figure 1: Discontinuity in Treatment Status at the Cutoff

RDD - Graphs: Zooming In Around the Cutoff



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RDD - Key Graph: Outcome against the Running Variable

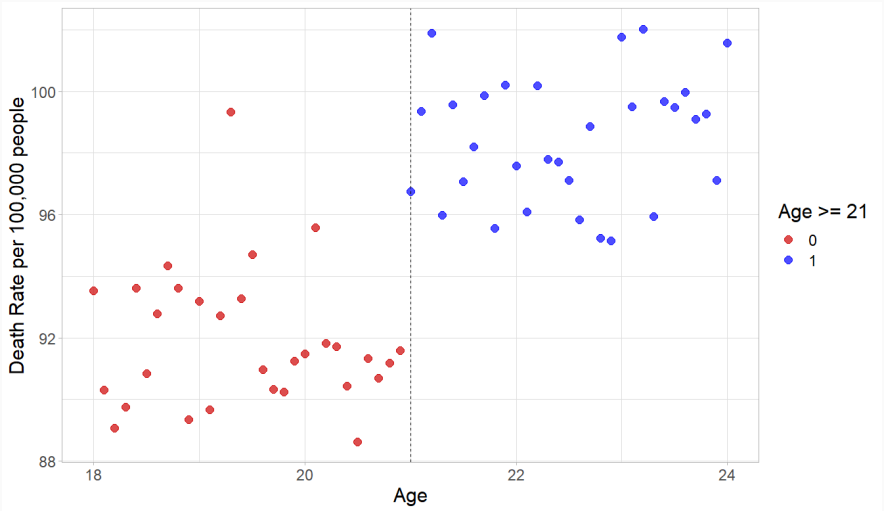


Figure 2: Discontinuity in the Outcome at the Cutoff

Estimation of the size of the gap

We can write the following simple regression model:

$$Deathrate_a = \alpha + \beta a + \delta D_a + u$$

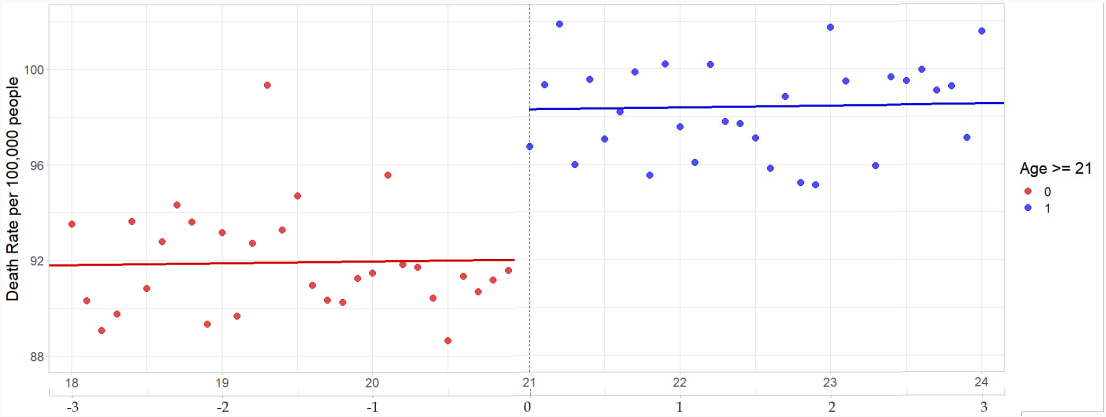
$Deathrate_a$ – death rate at age a,
a – age (defined in months relative to 21st birthday; Running Var.),
D_a – treatment dummy.

δ captures the jump in death rate between individuals above and below 21 years old.

The RDD estimator exploits a discontinuity at $a = 21$ in the conditional expectation function:

$$\underbrace{\lim_{c \to 21^+} \mathbb{E}[deathrate_a | a = c]}_{\alpha + \delta} - \underbrace{\lim_{c \to 21^-} \mathbb{E}[deathrate_a | a = c]}_{\alpha} = \delta$$

RDD - Graphical Results



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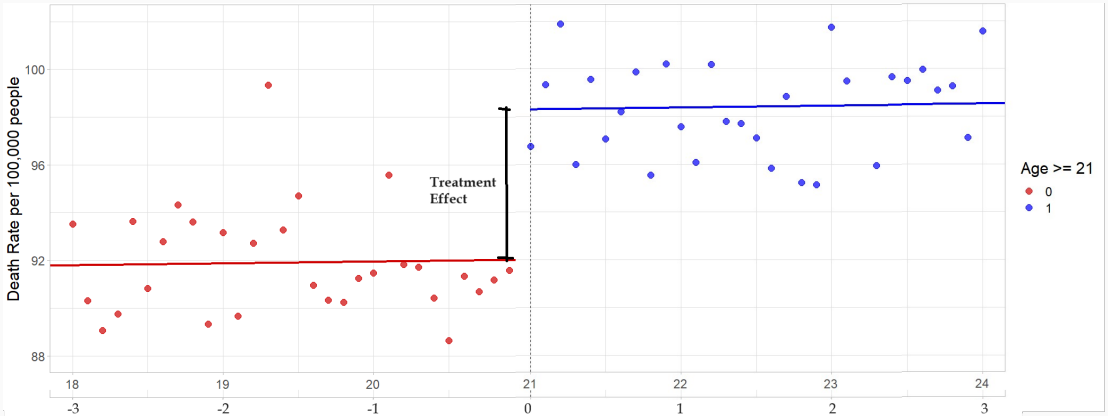
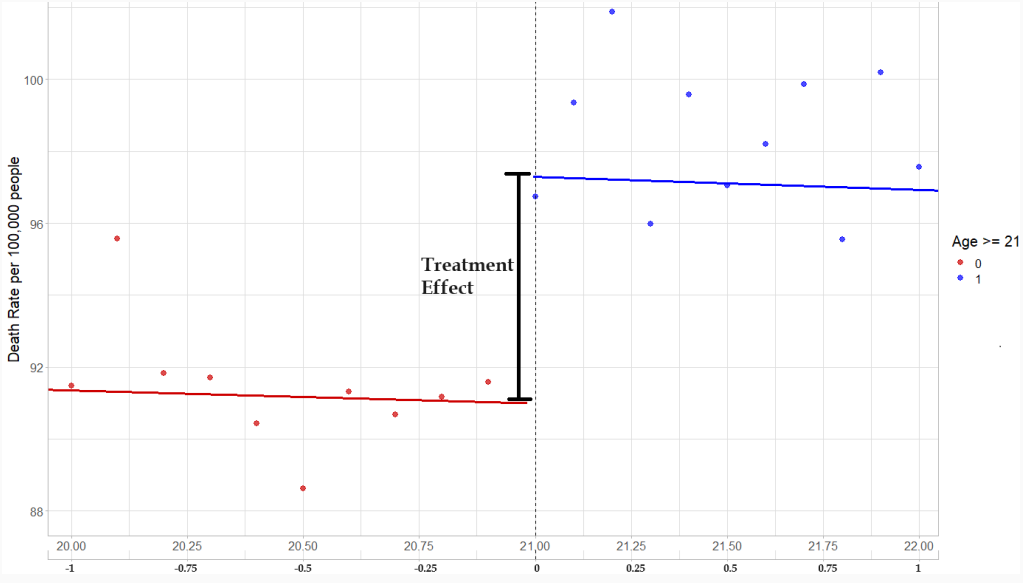


Figure 3: RDD Treatment Effect Estimate

RDD - Graphical Results



RDD - Regression

$$Deathrate_a = \alpha + \beta(a - cutoff) + \delta D_a + u$$

«include = F»= library(sandwich) library(lmtest) df <- read.csv("dataRDD.csv") df$age_c entered < -df$ age$_c$ell − 21@ <<>>= $mod1 <$ $-lm(death_r ate_p er_1 00 T\ age_c entered + treat, data = df)coeftest(mod1, vcov. =$ $vcovHC(mod1,"HC1"))$@

RDD: Details and Issues

To conduct an RD analysis, the researcher has to make several choices

1. Functional Form

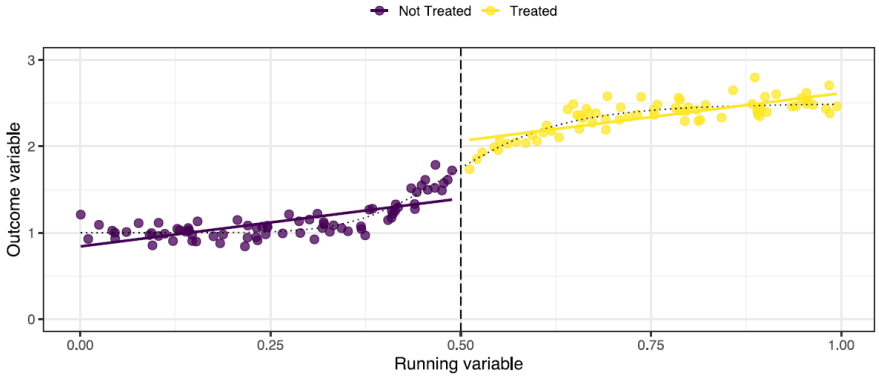
- the size of the gap depends on the type of lines you choose on each side of the cutoff
- insufficiently flexible specification: risk of mistaking nonlinearity for treatment effect
- overly flexible specification: reduces precision and runs the risk of overfitting.

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1. Functional Form

- the size of the gap depends on the type of lines you choose on each side of the cutoff
- insufficiently flexible specification: risk of mistaking nonlinearity for treatment effect
- overly flexible specification: reduces precision and runs the risk of overfitting.
- essential to visualise the data to choose an appropriate functional form!
- **Try different options $\rightarrow$ coefficients across models shouldn't vary too much if there is a "true" treatment effect.**
- estimators based on linear or quadratic polynomials or other smooth functions are recommended.

Simulations - Linear Relationship but NO Discontinuity



Souce: Oswald, Kenedi and Villedieu

Choosing the Functional Form

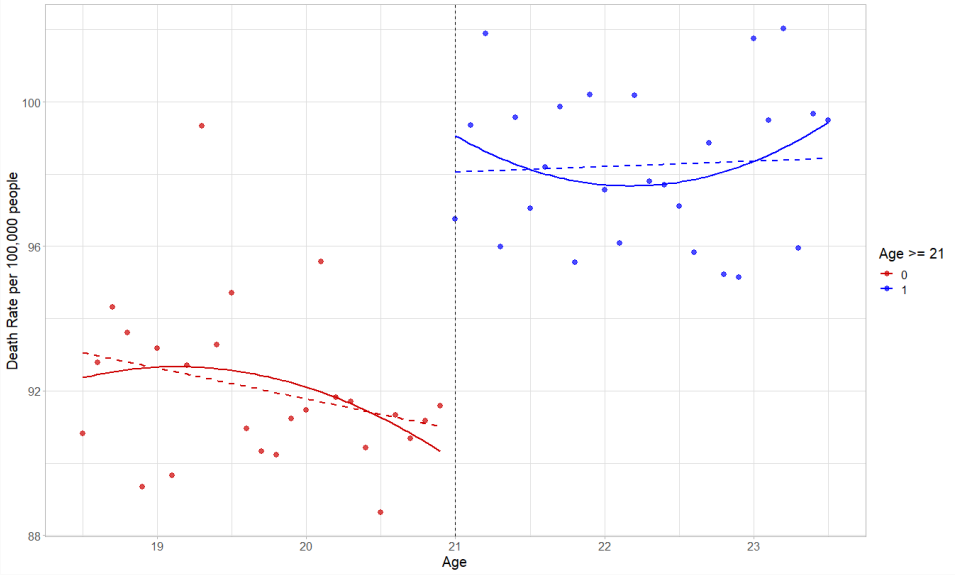
- Commonly, we allow the relationship between the outcome and the running variable to differ on the two sides of the discontinuity (varying slopes):

$$Y = \alpha + \beta(a - cutoff) + \gamma(a - cutoff) \cdot D_a + \delta D_a + u$$

- With a quadratic polynomial we have:

$$\begin{aligned} Y =& \alpha + \beta_1(a - cutoff) + \beta_1(a - cutoff)^2 + \\ & \gamma_1(a - cutoff) \cdot D_a + \gamma_2(a - cutoff)^2 \cdot D_a + \delta D_a + u \end{aligned}$$

Graphical Results with different Functional Forms



Regressions with Different Functional Forms

$$Deathrate_a = \alpha + \beta(a - cutoff) + \gamma(a - cutoff) \cdot D_a + \delta D_a + u$$

«»= mod2 <- lm(death$_r ate_p er_1 00 T$ $age_c entered * treat, data = df)coeftest(mod2, vcov. = vcovHC(mod2, "HC1"))@

Regressions with Different Functional Forms

$$Deathrate_a = \alpha + \beta_1(a - cutoff) + \beta_1(a - cutoff)^2 + \gamma_1(a - cutoff) \cdot D_a + \gamma_2(a - cutoff)^2 \cdot D_a + \delta D_a + u$$

$$\text{«»= mod3 <- lm(death}_r ate_p er_1 00 T \ age_c entered * treat + I(age_c entered^2) * treat, data = df)coeftest(mod3, vcov. = vcovHC(mod3, "HC1"))@$$

To conduct an RD analysis, the researcher has to make several choices

In our estimation, we want to give more weight to observations close to the cutoff level. There are two key parameters to do that:

2. Bandwidth

- **Bandwidth** = window around cutoff outside of which you discard observations (e.g. only consider individuals aged 19 to 23; or only those aged 20 to 22)
- bandwidth choice: comparability of observations around the cutoff vs. sample size

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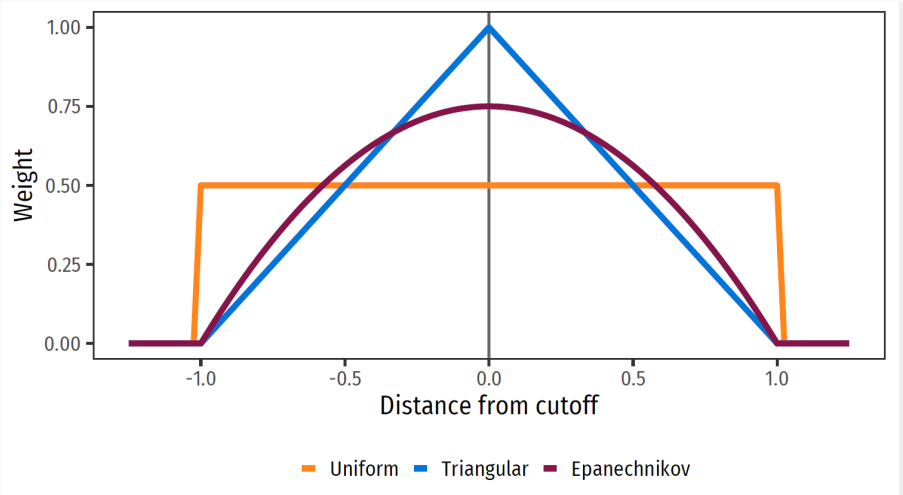
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3. Kernel

- **Kernel** = method for assigning importance to values based on their distance to the cutoff
- because we care the most about observations right by the cutoff, we typically want to give more distant ones less weight

Different Kernels



Source: Andrew Heiss

RDD - Issues

Your estimate of the treatment effect depends on all these:

- Functional form/line type
- Bandwidth (wide vs. narrow)
- Kernel weighting

Luckily there is an R package that chooses optimal bandwidth and kernel based on fancy algorithms: `rdrobust`.

For robustness, check what happens if you double/halve the bandwidth, use different kernels and different functional forms

Identification and Sources of Bias

Identifying Assumption

Key assumption: Potential outcomes are smooth at the threshold. In other words, in the absence of treatment, there would be no discontinuity.

- The population just below must not be discretely different from the population just above the cutoff.
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One thing we can do to "test" this assumption is a *placebo test*: Check if variables *other* than treatment or outcome vary at the cutoff (e.g. control variables)
If we find a significant discontinuity, that's bad! That tells us that other things are changing at the cutoff.

Checking for Manipulation: Density of the Running Variable

Manipulation of the running variable around the cutoff often generates a big cluster of observations to one side of the cutoff (bunching) and a gap on the other side

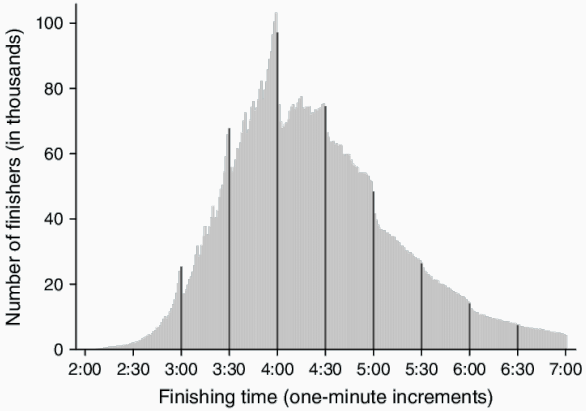


Figure 4: Distribution of Marathon Finishing Times

Another Issue: Noncompliance

Noncompliance

The running variable does not fully determine assignment to treatment. At the cutoff, your chance of treatment *increases* (jumps upward), but not from 0 to 1.

$\rightarrow$ **Fuzzy RDD** (instead of sharp RDD)

We will not cover the fuzzy RDD in detail, just know that the problem of imperfect determination of allocation to treatment can still be solved.

Practice in R

Effect of Training Program on Exam Score

If students score **below 40 points** (out of 100) on an initial test, they enter a special training program. We are interested in the effect of this program on the score in the final exam.

1. Open a new `.qmd` file and import the dataset *test_scores.csv*
2. Create a dummy variable for individuals scoring below 40 points on the initial test (*test_score*).
3. Plot the *final_score* as a function of the initial test score, colouring observations above and below 40 points. Choose a suitable bandwidth. Do you see a discontinuity?
4. Generate the same plot but with a re-centered test score variable that is 0 at the cutoff.

Effect of Training Program on Exam Score

5. Estimate the following model: $\textit{Final Score} = \alpha + \beta\,\textit{Test Score} + \delta\,\textit{Treated} + u$ (using a recentered test score variable).
   Does the RDD coefficient fit your graphical evidence?
6. How do you interpret each coefficient?
7. Estimate a model allowing for different linear slopes on each side of the cutoff. Then also estimate a model allowing for different quadratic specifications on each side. Do the RDD coefficients differ across models?
8. Generate the RDD plot using `rdplot()` from the package `rdrobust()`, and add regression lines.
9. Estimate the RDD coefficient using `rdrobust()`.